Adult Monomorphic Wide-Complex Tachycardia

AC-07 | Adult Cardiac

Quick flow

01 Confirm Pulse + Monomorphic Wide Rhythm [EMT / AEMT / Paramedic]

QRS ≥0.12 sec • assume VT when uncertain • airway/ventilation • oxygen if hypoxemic • monitor/pads • BP/SpO_2 • IV/IO

02 Instability Caused by Tachycardia? [EMT / AEMT / Paramedic]

Hypotension • acute altered mental status • shock • ischemic chest discomfort • acute heart failure

03 Unstable → Synchronized Cardioversion [Paramedic]

100 J biphasic • increase if unsuccessful • confirm sync before every shock • sedate when feasible

04 Stable + Uncertain Origin [Paramedic]

Only if regular and monomorphic: adenosine 6 mg rapid IV/IO + flush, then 12 mg after 1-2 min if needed; may repeat 12 mg once

05 Stable Monomorphic VT [Paramedic]

Amiodarone 150 mg IV/IO over 10 min • stop and cardiovert immediately if instability develops

06 Persistent After Infusion? [Paramedic]

Synchronized cardioversion 100 J biphasic • do not stack multiple antiarrhythmics

07 Converted / Recurrence [Paramedic]

If VT recurs: repeat amiodarone 150 mg over 10 min once • then infusion 1 mg/min

08 Reassess + Transport [EMT / AEMT / Paramedic]

Post-treatment rhythm strip/12-lead • complete vitals • early notification • prepare for polymorphic rhythm or pulseless arrest

Overview

- Adult pathway for patients 16 years of age and older with a pulse and a regular monomorphic wide-complex tachycardia, generally QRS duration of at least 0.12 seconds.
- When the diagnosis is uncertain, presume ventricular tachycardia and avoid medications that may worsen an undifferentiated wide-complex rhythm.
- Claiborne EMS uses an initial synchronized biphasic energy of 100 J for monomorphic ventricular tachycardia with a pulse.

EMT Care

Recognition, supportive care, pad placement, reversible-cause assessment, and immediate escalation for instability.

EMT

- Perform the primary assessment, confirm a pulse, support airway and ventilation, and give oxygen only when hypoxemic.
- Obtain complete vital signs, apply the cardiac monitor/AED and defibrillation pads, and obtain a 12-lead ECG when trained and equipped without delaying urgent treatment.
- Treat an uncertain wide-complex tachycardia as ventricular tachycardia and immediately request paramedic intervention.

- Identify possible ischemia, electrolyte abnormality, hyperkalemia, toxicologic exposure, stimulant use, or medication effect.
- Prepare for immediate CPR and defibrillation if the patient becomes pulseless or the rhythm becomes polymorphic.

AEMT Care

All EMT care plus vascular access, treatment preparation, and continuous reassessment.

AEMT

- Perform all EMT actions and establish IV/IO access without delaying synchronized cardioversion or transport.
- Acquire a 12-lead ECG before treatment when the patient is stable and assist with continuous rhythm recording during adenosine.
- Prepare medication, suction, ventilation equipment, and airway equipment for cardioversion and sedation.
- Reassess blood pressure, mental status, perfusion, chest discomfort, respiratory status, and rhythm after every intervention.
- Do not independently administer adenosine, amiodarone, or procedural sedation under AC-07.

Paramedic Care

All AEMT care plus rhythm interpretation, synchronized cardioversion, adenosine, amiodarone, and procedural sedation.

Paramedic

- Perform all EMT and AEMT actions; confirm rhythm regularity, monomorphic morphology, QRS duration, pulse, and whether the tachycardia is causing instability.
- For unstable monomorphic VT with a pulse, perform synchronized cardioversion at 100 J biphasic and increase energy if unsuccessful.
- Confirm synchronization markers before every shock and resynchronize after every attempt.
- Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.
- For a stable regular monomorphic wide-complex rhythm of uncertain origin, consider adenosine 6 mg rapid IV/IO with immediate flush, followed by 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once.
- For stable monomorphic VT, give amiodarone 150 mg IV/IO over 10 minutes; stop the infusion and cardiovert immediately if instability develops.
- After conversion, treat one recurrence with amiodarone 150 mg over 10 minutes, then begin maintenance infusion at 1 mg/min.

Indications

- Regular monomorphic wide-complex tachycardia with a pulse in a patient 16 years of age or older.
- Sustained monomorphic ventricular tachycardia requiring synchronized cardioversion or antiarrhythmic treatment.
- Stable regular monomorphic wide-complex tachycardia of uncertain origin in which adenosine may be diagnostic or therapeutic.

Contraindications

- Do not use adenosine for unstable, irregular, or polymorphic wide-complex tachycardia or active severe bronchospasm.
- Do not administer diltiazem, verapamil, metoprolol, or another AV-nodal blocker for an undifferentiated wide-complex rhythm.

- Do not administer amiodarone for suspected torsades de pointes, a clearly prolonged-QT polymorphic rhythm, or suspected pre-excited atrial fibrillation/flutter.

Assessment | Initial assessment

- Maintain the airway, assist ventilation as needed, and give oxygen only for hypoxemia.
- Obtain continuous cardiac monitoring, pads, frequent blood pressure, SpO_2, IV/IO access, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG before treatment when stable enough and it will not delay cardioversion.
- Determine onset, prior dysrhythmia, cardiac history, implanted devices, medications, stimulant or toxicologic exposure, and anticoagulant therapy.

Assessment | Instability attributable to tachycardia

- Hypotension, acutely altered mental status, shock or poor perfusion, ischemic chest discomfort, or acute heart failure.
- Confirm that the rhythm is driving instability rather than representing a secondary response to another critical condition.

Assessment | Reversible causes and rhythm hazards

- Evaluate for ACS, structural heart disease, electrolyte abnormality, hyperkalemia, hypomagnesemia, stimulant exposure, and medication toxicity.
- Consider sodium-channel-blocker toxicity with wide QRS, hypotension, seizures, or a compatible ingestion.
- If the rhythm becomes irregular, polymorphic, or pulseless, leave AC-07 immediately and open the linked pathway.

Treatment steps

- Treat an uncertain regular monomorphic wide-complex tachycardia as ventricular tachycardia and apply pads before deterioration.
- For instability attributable to monomorphic VT with a pulse, perform immediate synchronized cardioversion at 100 J biphasic and increase energy if unsuccessful.
- Confirm synchronization markers before every shock and resynchronize after every attempt.
- If synchronization is delayed and the condition is critical, deliver an unsynchronized high-energy shock.
- Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.
- For stable regular monomorphic wide-complex tachycardia of uncertain origin, consider adenosine 6 mg rapid IV/IO with immediate flush; then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once.
- Do not give adenosine for unstable, irregular, or polymorphic rhythm or active severe bronchospasm.
- For stable monomorphic VT, give amiodarone 150 mg IV/IO over 10 minutes. Stop the infusion and cardiovert immediately if instability develops.
- If stable VT persists after the infusion, perform synchronized cardioversion at 100 J rather than stacking antiarrhythmics.
- After conversion, if VT recurs, repeat amiodarone 150 mg over 10 minutes once, then begin maintenance infusion at 1 mg/min.
- Do not combine amiodarone with lidocaine or another antiarrhythmic under AC-07 unless directed by Medical Control.
- Obtain post-treatment rhythm strips and a 12-lead ECG, reassess complete vital signs, notify early, and transport.

Medications

Adenosine

6 mg rapid IV/IO with immediate flush; then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once

- Paramedic standing order only for stable regular monomorphic wide-complex tachycardia of uncertain origin.
- Use proximal access and record the rhythm continuously.
- Do not use for unstable, irregular, or polymorphic rhythm or active severe bronchospasm.

Amiodarone

150 mg IV/IO over 10 minutes

- Paramedic standing order for stable monomorphic VT with a pulse.
- Stop the infusion and cardiovert immediately if instability develops.
- After conversion, if VT recurs, repeat 150 mg over 10 minutes once and begin maintenance infusion at 1 mg/min.
- Do not combine with another antiarrhythmic without Medical Control.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic sedation for synchronized cardioversion when feasible.
- Do not delay cardioversion in an unstable patient.
- Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Warnings / precautions

- Treat an uncertain wide-complex tachycardia as ventricular tachycardia.
- Do not administer diltiazem or verapamil for undifferentiated wide-complex tachycardia.
- Adenosine is permitted only for stable regular monomorphic rhythm and is contraindicated in active severe bronchospasm.
- Do not use adenosine, amiodarone, diltiazem, metoprolol, digoxin, or another AV-nodal blocker for suspected pre-excited atrial fibrillation/flutter.
- Do not use amiodarone for torsades de pointes or a clearly prolonged-QT polymorphic rhythm.
- Do not stack antiarrhythmics without explicit Medical Control direction.
- Verify synchronization before every cardioversion attempt; many devices require reactivation after each shock.
- Sedation must not delay cardioversion. Loss of pulse requires immediate transition to AC-03 and AC-09.

Clinical pearls

- In an older adult or patient with structural heart disease, regular wide-complex tachycardia is VT until proven otherwise.
- Apply pads early because a stable patient may deteriorate rapidly.
- Adenosine may terminate SVT with aberrancy or reveal atrial activity; it does not treat ventricular tachycardia.
- A change to polymorphic morphology requires immediate transition to AC-08.
- Capture the rhythm before, during, and after medication or cardioversion whenever feasible.

Special populations | Pre-excited atrial fibrillation/flutter

- An irregular rapid wide-complex rhythm with variable morphology or known Wolff-Parkinson-White may represent pre-excitation.

- Avoid adenosine, amiodarone, and AV-nodal blockers; cardiovert if unstable and contact Medical Control early if stable.

Special populations | Toxicologic or metabolic wide-complex rhythm

- Hyperkalemia and sodium-channel-blocker toxicity require immediate cause-specific treatment in addition to rhythm and perfusion support.
- Open the linked renal/hyperkalemia or toxic-ingestion protocol rather than relying on amiodarone alone.

Related pathways / contacts

AC-04 ACS / STEMI - Open for ischemic chest discomfort or suspected ACS.

AC-08 Polymorphic Wide-Complex Tachycardia - Open immediately if the rhythm becomes polymorphic.

AC-09 VF / Pulseless VT - Open immediately if ventricular tachycardia becomes pulseless.

AC-03 Adult Cardiac Arrest - Open the master cardiac-arrest pathway after loss of pulse.

AM-03 Dialysis / Renal Failure - Open for suspected hyperkalemia or dialysis-related emergency.

TE-07 Overdose / Toxic Ingestion - Open for suspected sodium-channel-blocker or toxicologic dysrhythmia.

Source and references

American Heart Association. 2025 Adult Tachyarrhythmia With a Pulse Algorithm. | American Heart Association. 2025 Electrical Cardioversion Algorithm. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Advanced Life Support. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related cardiac, renal, and toxicology protocols.

Revision: October 16, 2026 | Last verified: October 16, 2026